

Ajay Venkatesh

Brooklyn, NY

+1 516 585 9013

av3855@nyu.edu

May 19, 2026

Hiring Manager

Paces

Dear Hiring Manager,

My name is Ajay Venkatesh. I'm based here in Brooklyn, finishing my M.S. in Computer Engineering at NYU, with several years of production software experience: a few years at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. My focus has been on full-stack product engineering across **TypeScript**, **Python**, **SQL**, the data layer underneath, and the practical **AI agent** layer that automates real human-driven workflows.

The Paces posting maps closely onto how I already work. At Amazon I deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details and generate context-aware summaries and remediation steps. I also implemented a **Model Context Protocol (MCP) server** for tool-augmented retrieval, materially reducing AWS Athena query time and compute costs. At Campus Mesh I built **CampusIQ**, an LLM-powered AI assistant with **prompt engineering** and **RAG retrieval** over structured data, plus **embedding-based semantic search** for contextual accuracy. Deploying agents in production isn't a side experiment for me, it's been the shape of my last year of work.

On full-stack data-intensive engineering: at NYU I shipped an end-to-end insurance analytics platform on **React**, **Node.js**, **REST APIs**, deployed via **Docker** on **EC2**, with Stripe billing, role-based access, and row-level security delivered to multiple paying clients. I engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control, processing hundreds of thousands of records, and powered **AWS QuickSight** dashboards for real-time analytics. At PwC I spent three years on a distributed **Microsoft Azure** backend over **SQL Server**, with event-driven processing, fault-tolerant retries, and parallelized batch ingestion that cut execution from hours to minutes. Across all three I've owned features end-to-end through design, coding, testing, and deployment, often working from ambiguous starting specs.

Stack-wise, I write **TypeScript** and **Python** daily, am comfortable across **PostgreSQL**, **MySQL**, **MongoDB**, **DynamoDB**, and I'm familiar with **LangChain**, **LangGraph**, and modern agent SDKs from the LLM work above. I'm equally at home in product-team conversations about roadmap as I am debugging production incidents through logs, metrics, and traces.

What pulls me toward Paces specifically is the combination: a Brooklyn startup, agentic AI applied to a domain that actually matters, and a product space where the work isn't synthetic. Renewable infrastructure and grid modeling are problems where being correct compounds into real-world climate outcomes. That's the kind of feedback loop I want to be inside of, alongside a small team where ownership is real.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh